ASE CHAPTER 1

V1 · Lecture-1 Vocabulary Coverage

The instructor's Lecture-1 terms mapped to their coverage — English terms, brief Arabic notes

Supporting-source answers for the two lecture gaps

11 · How are error, fault, system error, and failure related?

> 📝 **Answer**
>
> **A human error or mistake can introduce a system fault. When faulty code runs, it can create a system error, an incorrect system state. If the system then fails to provide the expected service, a system failure occurs. In software-process usage, defect and fault are synonyms, not two separate stages.**
>
> **Sommerville, §11.2, p.297, Fig. 11.3; Pressman, §8.4.1, p.203.**

12–13 · What does professional software ethics require, and which code covers it?

> 📝 **Answer**
>
> **Software engineers have responsibilities beyond technical work and should act ethically and responsibly. The joint Software Engineering Code of Ethics and Professional Practice has eight principles: Public, Client and Employer, Product, Judgment, Management, Profession, Colleagues, and Self.**
>
> **Sommerville, §1.2, pp.15–16.**

Rapid self-test — recall prompts (no answers shown)

1. Which two topics are missing from Aggarwal Chapter 1, and which supporting sources cover them? How is `defect` related to `fault`?
2. Where exactly are the IBM figures printed, and what are the three numbers?
3. Which failure cases are in the book's own list, and which one comes only from the slides?
4. What components does the source include in software?